

Can AI have consciousness?

“I think we are moving into a period when, for the first time ever, we may have things more intelligent than us,” said Geoffrey Hinton in a 2023 interview (Hinton, 2023). Hinton, who went on to win the 2024 Nobel Prize in Physics for his work in artificial intelligence, expressed his belief that Artificial Intelligence (AI) would eventually develop self-awareness and consciousness. Now, nearly a year and a half later, AI continues to advance at an unstoppable pace—yet the question of consciousness remains unresolved, and even top-level scientists like Hinton have not been able to convince everyone.

Before diving into this question, we first need to clarify what kind of consciousness people usually refer to. In our earlier homework, we briefly touched on this. David Chalmers categorized the study of consciousness into two types of problems (Wagner-Altendorf, 2024). The “easy problem” refers to understanding the physical mechanisms of the brain—how neurons wire, what regions control certain functions, and how signals are processed. However, as Stuart Hameroff and others emphasize, there remains the “hard problem” of consciousness: explaining subjective experience itself: why and how we feel, perceive, and have awareness (New Thinking Allowed with Jeffrey Mishlove, 2017). Many scientists have proposed theories to explain consciousness, including the Neural Correlates of Consciousness and Integrated Information Theory (IIT). In general, these theories fall into two camps—those that view consciousness as a purely computational process and those that do not.

When it comes to AI, the concept of consciousness becomes even more abstract. People typically argue that a computer is not conscious because it can’t “think” like a human being but only simply execute code or follow algorithms. For example, when a person thinks about the equation $1 + 1$, the brain just seems to arrive at the answer “2”. In fact we can’t completely understand how that mental process happens during the calculation. In contrast, a computer without consciousness produces the same answer based purely on predefined logic. It isn’t thinking like a human; it’s just calculating and executing. Right now what we can be sure of is that AI has intelligence to handle complex problems better than most

humans. However, does it truly have feelings? Perhaps one day, we'll feel compelled to thank AI after each use, just to avoid the risk of future revenge.

A Large Language Model (LLM) is a type of artificial intelligence trained on massive amounts of text data to generate human-like responses in natural language. I choose to narrow the scope of this discussion from AI in general to LLMs because they currently represent the closest technological approach to the path of obtaining consciousness. They sparked growing concerns about conscious AI becoming uncontrollable. At this point, one thing we can confidently say is that LLMs are capable of pretending to have consciousness. If someone interacts with an LLM using the right prompt, the conversation can feel remarkably human-like, making it difficult to distinguish the model from a real person.

But an LLM engineer may tell you that what people are observing is not consciousness, but reasoning. Reasoning refers to the model's ability to draw conclusions, infer relationships, or generate logical responses based on patterns it has learned from data in the context of LLMs (Cosentino & Shekkizhar, 2024). It mimics human-like thought by following probabilistic rules and context clues embedded in the language it has processed. Because of this, an LLM can provide answers that appear thoughtful or intentional especially when prompted well. It can explain, summarize, argue, and even reflect tone or emotion to create the illusion of a conscious mind. However, this is not real consciousness. The LLM does not know what it is saying. It has no sense of understanding. Its outputs are the result of statistical correlations. From the code level, an LLM just operates by assigning probabilities to sequences of words using large matrices of numerical weights.

Based on the earlier discussion about code, some people believe that as AI systems grow more complex, consciousness will eventually emerge as a natural consequence of evolution. From this point, consciousness is expected to arise spontaneously from increasingly sophisticated code. I would categorize this perspective as "ultimate optimism." This type of optimism is closely related to another positive viewpoint: materialism. According to materialism, the brain—which gives rise to consciousness—is simply a complex arrangement of physical materials found in the real world. In theory, this suggests that

AI could one day become conscious, since we might eventually build a computer or system that replicates the human brain at the atomic level.

Kyle Fish, a young researcher at Anthropic who has extensively studied the intersection of AI and consciousness, supports this possibility. He notes that modern AI systems are evolving rapidly and are no longer limited to being housed within a single computer. They can be integrated with sensory technologies such as cameras, radar, and other tools to provide them with experiences including touch, smell, and taste (Glover, 2025). By combining all this multimodal data, these systems may develop a more holistic and comprehensive understanding of the world, potentially leading to an interesting sense of self in the future.

Another positive viewpoint differs slightly from materialism and is more closely aligned with panpsychism. Panpsychism generally suggests that everything in the universe possesses some form of consciousness or spirit—even something as simple as a rock. However, this idea is not meant to be taken as a religious statement. IIT, which I mentioned earlier, supports this perspective. Dr. Giulio Tononi introduced IIT as a way to understand human consciousness as the integration of information, measured by a threshold value called Φ (phi). According to the theory, humans have consciousness because our brains, when active, exhibit strong integration between different regions processing sensory input, resulting in a high Φ . Objects or animals can also possess consciousness, although their Φ may be super lower, indicating a weaker form of awareness. So how does this relate to AI consciousness? In systems like LLMs, we already have large networks of artificial neurons processing vast amounts of information. This suggests that they may have some degree of Φ —we just don't yet know whether that level is sufficient for consciousness to arise. Technically, it's not difficult to increase Φ to a great level, since we can manually design more complex systems with greater interconnectivity.

Another viewpoint supports the idea that AI can have consciousness, though I believe it has certain unsolved problems. This theory is one of the most debated today. It's called Predictive Processing Theory (PP), also known as Predictive Coding Theory. It suggests that consciousness is essentially a form of hallucination. Anil Seth, a British cognitive scientist, describes the brain as engaging in controlled

hallucinations (Seth, 2021). For example, when we see a red painting, the feeling of seeing "red" arises not directly from the painting, but because our brain maintains an internal model that interprets sensory input such as light and environment and predicts the perception of red. As a result, we constantly compare incoming data with internal predictions, and adjust our model based on new experiences. This is known as the “inside-out” approach. The idea has roots in 18th-century German philosophy. Immanuel Kant argued that our perception of the external world is shaped to fit our cognitive framework (Brook & Wuerth, 2023). This ongoing adjustment of internal models is a form of Bayesian inference. Interestingly, this process is strikingly similar to how neural networks operate today. If this theory holds, then consciousness may not be a deep mystery—it could be replicable in machines through sufficiently advanced models. However, this theory still doesn't explain why there is a subjective experience behind the prediction. A machine might process information in the same way, but there's no evidence that it feels anything and it simply computes. But this brings us to the hardest part of the problem: if AI truly has consciousness, how would we know? There is currently no proven method to detect or measure consciousness, making the question nearly impossible to answer with certainty.

Roger Penrose, a renowned physicist and mathematician, offers one of the most influential quantum-based theories of consciousness. He argues that consciousness cannot be explained by classical computation alone and must involve quantum processes. Specifically, phenomena like wave function collapse at the microscopic level within the brain, possibly occurring in brain microtubules. Unlike binary code, which operates on 0s and 1s, quantum systems allow for uncertainty, superposition, and retrocausality, which Penrose believes are essential to creating consciousness. This gives some hope that AI can have consciousness if future large-scale quantum computers can be built and utilized. But I am pessimistic about this because people apparently are still far from deeply understanding either consciousness or quantum computers.

In a recent interview, Penrose emphasizes that based on current knowledge, AI cannot replicate the depth of subjective experience that human processes even though it can perform better than humans in some tasks (Okemwa, 2025). One of the key reasons for this claim is correlated to Gödel's

incompleteness theorem, which shows that there are truths in mathematics that cannot be proven by any algorithm. Penrose argues that human thought can operate beyond what is algorithmically possible. In other words, AI needs something more.

In conclusion, there are too many competing theories to give a simple answer to whether AI can have consciousness. Even before we tackle that question, we still don't have a unified definition of what consciousness actually is. Will this question ever be answered? Personally, I remain neutral. It's possible that no theory alone will be able to prove it definitively. People are so eager to find an answer because the question speaks to two deeper concerns: first, whether human beings are truly "special" with uniquely free and complex minds; and second, whether AI could one day break free from its computational limits and pose a real threat to humanity. If AI ever does break out of its constraints and begins acting in unpredictable, self-driven ways, perhaps the answer will reveal itself. Or maybe, like the mystery of what happens after death, this will remain a question we can never fully resolve.

References

- Cosentino, R., & Shekkizhar, S. (2024). Reasoning in large language models: A geometric perspective.
- 60 Minutes (Producer), & . (2023, Oct 10). *Geoffrey hinton: In time, AI will have self-awareness, consciousness*, [Video/DVD]
- Glover, E. (2025, May 02). Could AI become conscious? exploring the line between science and fiction.
- New Thinking Allowed with Jeffrey Mishlove (Producer), & Jeffrey Mishlove (Director). (2017, Aug 13). *The hard problem of consciousness with stuart hameroff*, [Video/DVD]
- Okemwa, K. (2025, Mar 26). Nobel laureate claims "AI will not be conscious" and shouldn't be considered intelligent — until it develops its own ideas.
- Seth, A. (2021). *Being you : A new science of consciousness*
- Wagner-Altendorf, T. A. (2024). Progress in understanding consciousness? easy and hard problems, and philosophical and empirical perspectives. *Acta Analytica*, 39(4), 719.
doi:10.1007/s12136-024-00584-5